

Customer Churn Prediction System

Software Requirements Specification — v1.0

1. Project Overview

The Customer Churn Prediction System shall enable the business to identify customers at risk of cancelling their subscription before it occurs. The system must ingest customer behaviour data, train and serve ML models, and expose predictions through a REST API and an analytics dashboard. The application should support batch scoring of the full customer base as well as real-time scoring for individual customers.

2. Functional Requirements

2.1 Data Ingestion

FR-01 — The system shall ingest customer data from CSV, JSON, and database sources on a configurable schedule.

FR-02 — The system must validate incoming data and reject records with missing required fields, logging each rejection.

FR-03 — The system shall support incremental ingestion so only new or updated records are processed on each run.

FR-04 — The system must allow users to upload a CSV file manually through the dashboard for ad-hoc analysis.

FR-05 — The system should archive raw ingested files for a minimum of 90 days for audit purposes.

2.2 Feature Engineering

FR-06 — The system shall compute recency, frequency, and monetary (RFM) features for each customer automatically.

FR-07 — The system must calculate rolling averages for login frequency, support ticket count, and purchase value over 7, 30, and 90-day windows.

FR-08 — The system shall encode categorical variables such as subscription tier, region, and payment method using one-hot or target encoding.

FR-09 — The system must handle missing values through imputation strategies configurable per feature column.

FR-10 — The system should detect and flag outliers in numerical features before model training.

2.3 Model Training

FR-11 — The system shall train a churn classification model using at least one of: Logistic Regression, Random Forest, or XGBoost.

FR-12 — The system must perform hyperparameter tuning using cross-validation and report the best parameters.

FR-13 — The system shall split data into train, validation, and test sets with configurable ratios (default 70/15/15).

FR-14 — The system must log training metrics including AUC-ROC, precision, recall, F1-score, and accuracy to an experiment tracker.

FR-15 — The system shall support retraining on a schedule or when triggered manually by an administrator.

FR-16 — The system must version each trained model and allow rollback to any previous model version.

2.4 Prediction & Scoring

FR-17 — The system shall expose a REST API endpoint that returns a churn probability score between 0 and 1 for a given customer ID.

FR-18 — The system must support batch scoring that processes the entire customer base and stores results in the database.

FR-19 — The system shall classify customers into risk tiers: Low (0–0.3), Medium (0.3–0.7), and High (0.7–1.0) based on their churn score.

FR-20 — The API must respond to individual prediction requests within 200 milliseconds at the 95th percentile under normal load.

FR-21 — The system should provide an explanation for each prediction, highlighting the top 3 features contributing to the churn score.

2.5 Dashboard & Reporting

FR-22 — The system shall provide a web dashboard that displays the overall churn rate trend over the last 12 months.

FR-23 — The dashboard must allow users to filter customers by churn risk tier, subscription plan, and region.

FR-24 — The system shall allow users to export a list of high-risk customers to CSV with their contact information and churn scores.

FR-25 — The dashboard must display model performance metrics and update them after each retraining cycle.

FR-26 — The system should send automated weekly email reports to configured recipients summarising churn risk statistics.

2.6 Alerting

FR-27 — The system shall trigger an alert when a customer's churn probability crosses the High threshold within a 24-hour window.

FR-28 — The system must allow administrators to configure alert thresholds and notification channels (email, Slack, webhook).

FR-29 — The system should suppress duplicate alerts for the same customer within a configurable cooldown period.

3. Non-Functional Requirements

3.1 Performance

NFR-01 — The batch scoring pipeline must process 1 million customer records in under 30 minutes.

NFR-02 — The REST API must handle at least 500 concurrent prediction requests without degradation.

NFR-03 — The dashboard must load and render within 3 seconds for datasets up to 100,000 customers.

3.2 Reliability & Availability

NFR-04 — The prediction API should maintain 99.5% uptime measured monthly.

NFR-05 — The system must retry failed ingestion jobs up to 3 times with exponential back-off before raising an alert.

NFR-06 — The system shall store all predictions durably so they are recoverable after a system restart.

3.3 Security & Privacy

NFR-07 — The system must authenticate all API requests using API keys or OAuth 2.0 tokens.

NFR-08 — Personally identifiable information (PII) such as customer email and phone number must be encrypted at rest.

NFR-09 — The system shall maintain an audit log of all data access, model training events, and configuration changes.

NFR-10 — The system must comply with GDPR and allow customer data to be deleted on request within 30 days.

3.4 Scalability & Maintainability

NFR-11 — The system should be containerised with Docker and deployable on Kubernetes for horizontal scaling.

NFR-12 — The codebase must have at least 80% unit test coverage for the feature engineering and scoring modules.

NFR-13 — New data sources should be integrable by implementing a standard ingestion interface without modifying core pipeline code.

4. Data Requirements

DR-01 — The system must accept a minimum customer dataset containing: customer_id, signup_date, last_login_date, total_purchases, monthly_spend, support_tickets_last_90d, subscription_tier, and churned label.

DR-02 — The training dataset must contain at least 5,000 labelled customer records with a churn rate between 5% and 40% to be considered valid.

DR-03 — The system shall detect and report class imbalance and apply oversampling (SMOTE) or class weighting when the churn rate is below 10%.

DR-04 — All date fields must be stored and processed in ISO 8601 UTC format.

5. Constraints & Assumptions

- The initial implementation targets Python 3.10+ and a scikit-learn / XGBoost stack.
- The system assumes customer IDs are stable and unique across all data sources.
- The MVP scope excludes deep-learning models; they may be added in a future release.
- The system is assumed to run in a private cloud environment with access to the customer database via an internal network.